

INDUCTION OF SEXUAL RECEPTIVITY IN ESTROGEN CONDITIONED SPAYED FEMALE GUINEA PIGS BY ORALLY ADMINISTERED PROGESTERONE AND PREGNENINOLONE*

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PREVIOUS INVESTIGATIONS of the effectiveness of orally administered progesterone and progesterone-like compounds have demonstrated the necessity of larger doses if results comparable to those produced by subcutaneous injection are to be obtained. In the rabbit the minimal oral dose of corpus luteum hormone necessary to produce distinct progestational proliferation is 50 rabbit units.¹ Even this amount did not produce as marked an effect as 1 rb. u. injected subcutaneously. Hohlweg and Inhoffen² were unable to detect any alterations in the uteri of rabbits when as much as 60 mg. progesterone was given orally over a 5-day period. This amount represents 100 times the effective subcutaneous dose. On the other hand, they were successful in producing progestational changes with orally administered pregneninolone in quantities (4 mg.) twice that necessary by injection. Clauberg and Ustün³ also report that pregneninolone (proluton-C) is effective when given orally to humans in 6 times the intramuscular dosage.

TABLE 1. INDUCTION OF RECEPTIVITY BY ORALLY ADMINISTERED PROGESTERONE AND PREGNENINOLONE

No. of animals	Quantity	No. in heat	% in heat	Latent period		Length of heat		Combined latent and length
				Mean	Range	Mean	Range	
Progesterone								
	mg.							
3	15.0	3	100.0	3.0	2.5- 3.5	8.6	6.5-10.0	11.6
2	10.0	2	100.0	3.8	3.5- 4.0	6.5	5.5- 7.5	10.3
3	5.0	3	100.0	3.5	3.5- 3.5	7.7	7.5- 8.0	11.2
3	4.0	3	100.0	3.7	3.5- 4.0	8.5	6.5- 9.5	12.0
4	2.0	4	100.0	4.4	4.0- 5.0	8.4	7.5- 9.5	12.8
14	1.5	14	100.0	4.6	3.5- 5.5	7.0	4.5- 8.5	11.6
2	1.4	2	100.0	4.8	4.0- 5.5	4.3	2.0- 6.5	9.1
7	1.3	7	100.0	4.4	4.0- 5.5	4.4	4.0- 5.5	8.8
10	1.2	9	90.0	5.3	3.5- 7.5	6.1	4.5- 8.5	11.4
7	1.1	6	85.7	4.8	3.0- 8.0	7.2	4.5-10.0	12.0
13	1.0	11	84.6	3.8	2.0- 6.0	7.7	6.0-11.0	11.5
11	0.8	6	54.5	4.3	2.5- 5.5	3.6	1.5- 6.5	7.9
9	0.6	5	55.5	4.4	3.0- 6.0	4.4	3.5- 5.5	8.8
8	0.4	3	37.5	4.3	3.5- 6.0	3.0	2.0- 3.5	7.3
18	0.2	3	16.7	4.8	4.0- 5.0	5.0	3.0- 7.5	9.8
Pregneninolone								
8	2.0	8	100.0	4.2	3.8- 5.0	9.4	6.8-10.8	13.6
16	1.0	10	62.5	5.0	2.5-11.0	6.1	2.5-10.0	11.1
Control injections of progesterone								
7	0.05	6	85.9	7.8	5.0-12.0	5.8	1.5- 8.0	13.6
9	0.1	8	88.8	4.9	4.0- 6.3	6.9	3.0-10.0	11.8

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¹ PORTMAN, K.: *Acta Path. et Microbiol. Scand., Suppl.* 25: 9. 1935.

² HOHLWEG, W., AND H. H. INHOFFEN: *Klin. Wchnschr.* 18: 77. 1939.

³ CLAUBERG, C. AND Z. USTÜN: *Zentralbl. f. Gynäk.* 62: 1745. 1938.

An explanation for the refractiveness of the animal organism to orally administered corpus luteum extract has been advanced by Engelhart⁴ who showed that the effective hormone is split in vitro and rendered ineffective by the pancreatic enzyme, lipase. This applies particularly to the corpus luteum extract whereas the purified crystalline hormone, progesterone, is more resistant.²

Using the induction of sexual receptivity in spayed female guinea pigs as a criterion of effectiveness,⁵ the action of progesterone (proluton, Schering)[†] given in capsules and pregnenolone tablets administered orally has been studied. The animals were observed at half-hourly or hourly intervals to determine the length of the latent period (interval between feeding and beginning of heat reaction) and heat.

The effects of graded doses of progesterone and pregnenolone are summarized in table 1. It is apparent that 1 mg. of progesterone was sufficient to induce normal copulatory behavior in a high percentage of animals previously conditioned with subcutaneously injected estrogen. Although lower doses were also effective, the percentage of animals responding was smaller. The response which followed administration of the smallest quantity, 0.2 mg., could have been caused by the injected estrogen. Pregnenolone was effective in 8 of 8 animals when 2 mg. were given and in 10 of 16 animals when 1 mg. was given.

Inasmuch as 0.05 mg. progesterone injected subcutaneously is sufficient to induce heat in a large percentage of animals,⁵ it appears that when the induction of heat in the female guinea pig is the end-point, the ratio of orally administered to injected progesterone is about 20:1. This ratio is considerably lower than that of 50:1 obtained by Portman¹ when pregestational changes in the rabbit uterus were used as the end-point.

⁴ ENGELHART, E.: *Arch. f. Gynäk.* 148: 76. 1938.

⁵ YOUNG, W. C., E. W. DEMPSEY, C. W. HAGQUIST AND J. L. BOLING: *J. Lab. & Clin. Med.* 23: 300. 1937.

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⁶ HERTZ, R., R. K. MEYER AND M. A. SPIELMAN: *Endocrinology* 21: 533. 1937.



ACETYLCHOLINE-EQUIVALENT CONTENT OF THE VAGINA AND VULVA OF OVARIECTOMIZED RABBITS, BEFORE AND AFTER ADMINISTRATION OF ESTROGEN*

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THE FOLLOWING DATA were obtained in connection with the observation that estrogen increases appreciably the concentration of free acetylcholine in the uterus of rabbits^{1,2} and in the nasal mucosa of ovariectomized rabbits and cats.³ The method of extraction and testing the tissues for this substance was the same as in earlier experiments; it was, namely, that of Chang and Gaddum,⁴ modified slightly as described elsewhere.¹ The rabbits were ovariectomized well in advance of the experiment, usually 18 to 21 days previously. At the time of the experiment, two rabbits were anesthetized with dial (Ciba) given intramuscularly. The tissues of one were taken without further treatment, those of the other, at the end of one hour after subcutaneous injection of estrogen (500γ of estradiol-dipropionate,

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¹ REYNOLDS, S. R. M.: *J. Physiol.* 95: 258. 1939.

² REYNOLDS, S. R. M., AND F. I. FOSTER: *Am. J. Physiol.* 128: 147. 1939.

³ REYNOLDS, S. R. M., AND F. I. FOSTER: *Am. J. Physiol.* In press.

⁴ CHANG, H. C. AND J. H. GADDUM: *J. Physiol.* 79: 255. 1933.